

Monel K500 | High-Strength & Corrosion-Resistant Nickel-Copper Alloy

Introduction:

Monel K500 (UNS N05500) is a precipitation-hardenable nickel-copper alloy combining the corrosion resistance of Monel 400 with superior strength and hardness via aluminum and titanium content. Non-magnetic and dimensionally stable, it excels in marine, chemical, oil & gas, and high-performance industrial applications.

Chemical Composition:

Element	Composition (%)
Nickel	63.0 min
Copper	27.0 - 33.0
Aluminum	2.3 - 3.15
Titanium	0.35 - 0.85
Iron	2.0 max
Manganese	1.5 max
Silicon	0.5 max
Carbon	0.25 max
Sulfur	0.01 max

Physical & Mechanical Properties:

- **Density:** 8.44 g/cm³
- **Melting Range:** 1315-1350 °C
- **Tensile Strength:** 130-220 ksi
- **Yield Strength:** 85-200 ksi
- **Elongation:** 3-35%
- **Hardness:** 24-40 HRC

Available Products:

- Seamless & Welded Pipes (ERW & EFW)
- Tubes (Capillary, U-Bend)
- Sheets, Plates, Coils & Strips
- Round, Flat, Hex & Square Bars
- Pipe Fittings, Flanges & Fasteners
- Custom Fabricated Components

Applications:

- Marine & Offshore Components (shafts, seawater valves, fasteners)
- Oil & Gas Equipment (downhole tools, drill collars, valve parts)
- Chemical Processing (pumps, impellers, heat exchangers)
- Aerospace & Defense (springs, fasteners, actuators)
- Power Generation (turbines, condensers, bolting)
- Medical & Electronics (precision, non-magnetic components)

Advantages:

- Superior corrosion resistance in seawater and chemicals
- High strength and hardness via precipitation hardening
- Non-magnetic at sub-zero temperatures
- Resistant to stress corrosion cracking and erosion
- Long service life with low maintenance

Why Choose Moksh Tubes & Fittings LLP:

- ISO-certified supplier with global reach
- Specialized stock and custom fabrication support
- Rapid delivery and certified quality materials
- Trusted by marine, chemical, oil & gas, and aerospace industries